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AI Conference

Learning to Adapt Retrieval-Augmented LLMs through Question Complexity

Somantha Manuranga | OTH Amberg-Weiden

Outline

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1. Introduction and Motivation
2. Methods and Experiments
3. Results and Discussion
4. Summary and Conclusion

Introduction and Motivation

What is a Large Language Model?

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- A type of language model designed to understand and generate human language
- Built using deep learning, especially a **transformer** architecture
- Trained on massive amounts of text data (books, websites, articles, code, etc.)

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What LLMs Don't Know

- **Private databases:** LLMs cannot access confidential information
- **Hard to access information:** some information is not widely available online
- **Real-time data:** LLMs are trained on past data and do not update automatically

Two Steps of Question Answering

Complexity determines the strategy

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Common knowledge:
"Why are hotels expensive on weekends?"

(1) Collect Information → (2) Reason & Respond

No retrieval needed

LLMs internal knowledge

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Up to Date Information:
"Why are Munich hotels pricier this weekend?"

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Extensive search needed — Synthesize retrieved docs

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How LLMs generate upto date information ?

Retrieval Augmented Generation (RAG)

What is RAG and why do we need it?

Izacard et al., JMLR 2023

Retrieval Augmented Generation (RAG)

What is RAG and why do we need it?

- RAG leverages **external knowledge** to improve LLM answer quality
- Simply by injecting retrieved documents into the prompt

Izacard et al., JMLR 2023

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User Query

Izacard et al., JMLR 2023

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Without RAG

LLM relies only on parametric memory

→ outdated or wrong

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With RAG

LLM gets fresh external knowledge

→ accurate & current

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The Answer

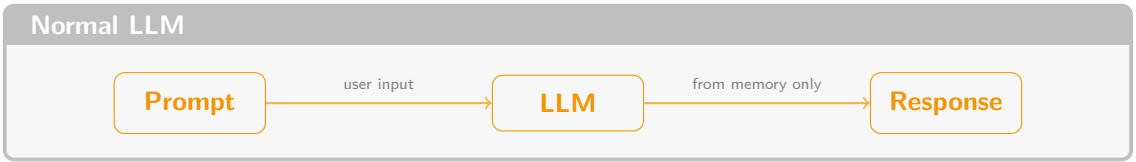
Just put retrieved docs in the prompt

→ no retraining needed

Izacard et al., JMLR 2023

Normal LLM vs. RAG System

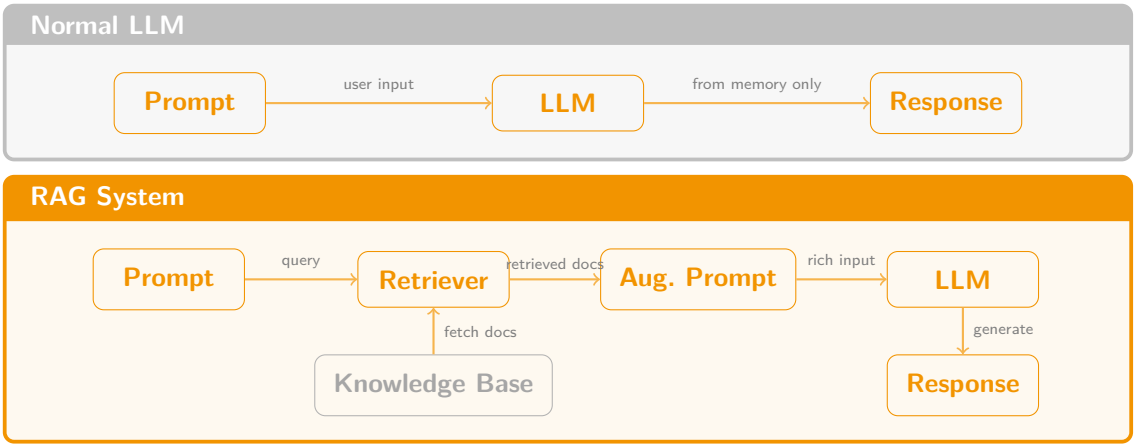
How retrieval augments the generation pipeline



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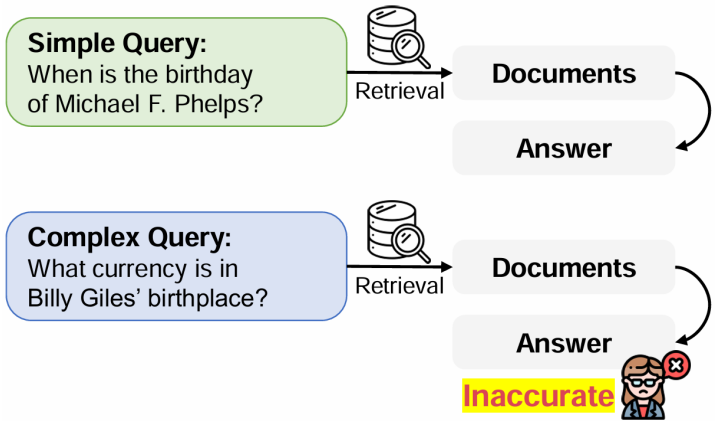
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Single Step Approach

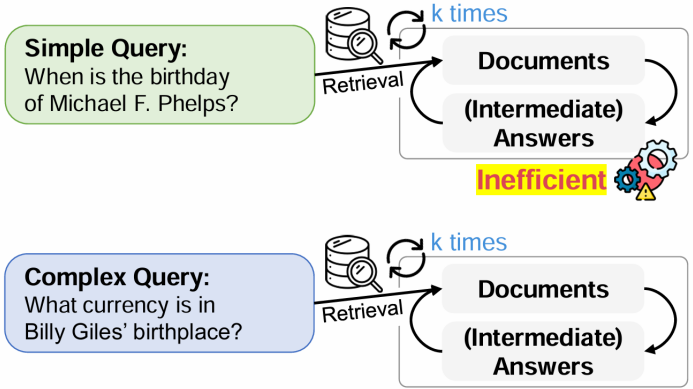
One retrieval for every query



Single retrieval works for simple queries.
fails *when multi-hop reasoning is needed.*

Multi-Step Approach

Iterative retrieval for every query

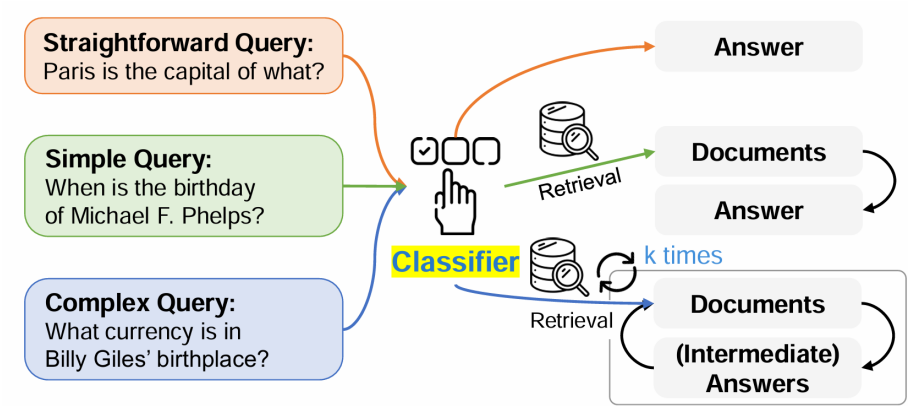


Works with complex queries.
Inefficient *for simple queries.*

Trivedi et al., ACL 2023

Adaptive-RAG (Ours)

A classifier routes each query to the right strategy



Right strategy, Right query

Formal Problem Definition

Three strategies - one unified framework

Mallen et al., ACL 2023

Formal Problem Definition

Three strategies - one unified framework

Label A No Retrieval

LLM answers from parametric memory only

$$\bar{a} = \text{LLM}(q)$$

q = input query

$\bar{a}$ = generated answer

Mallen et al., ACL 2023

Formal Problem Definition

Three strategies - one unified framework

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Label B Single-Step

Retrieve once, then augment the prompt

$$d = \text{Retriever}(q;\, D)$$

$$\bar{a} = \text{LLM}(q,\, d)$$

D = knowledge corpus

d = retrieved document

Mallen et al., ACL 2023

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Label C Multi-Step

Iterative retrieval over multiple reasoning steps

$$d_i = \text{Retriever}(q,\ c_i;\ D)$$

$$\bar{a}_i = \text{LLM}(q,\ d_i,\ c_i)$$

c_i = prior docs & intermediate answers

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Adaptive-RAG Classifier

$o = \text{Classifier}(q)$ where $o \in \{\mathbf{A},\ \mathbf{B},\ \mathbf{C}\}$ Selects the right strategy per query

Mallen et al., ACL 2023

Methods & Experiments

Experimental Setup

Datasets (6 Total)

Single-hop (Tier A/B)

SQuAD v1.1	Reading comprehension
Natural Questions	Real Google search queries
TriviaQA	Trivia questions from web

Multi-hop (Tier C)

MuSiQue	2–4 hop compositions
HotpotQA	Wikipedia multi-doc
2WikiMultiHopQA	2-hop Wikipedia

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LLMs Tested

FLAN-T5-XL	3B parameters
FLAN-T5-XXL	11B parameters
GPT-3.5 Turbo	Instruct variant

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Evaluation Metrics

F1 / EM / Accuracy	Task performance
Steps · Time/Query	Efficiency

- **Data:** T5 Large with 400 queries (from 6 datasets)
- **Optimization:** Cross-Entropy & AdamW

Query Complexity Classifier

Labeling Strategy

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Case 1: Silver Labels from Model Predictions

- Run all three strategies on each query
- **No-Retrieval** correct → label `'A'` *(Straightforward)*
- No-Retrieval fails, others correct → label `'B'` *(Moderate)*
- Only **Multi-step** correct → label `'C'` *(Complex)*
- Tie-breaking: **simpler model always gets priority**

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Case 2: Inductive Bias from Datasets

- Applied to queries **unlabeled** after Case 1
- Single-hop datasets (SQuAD, NQ, TriviaQA) → label `'B'`
- Multi-hop datasets (MuSiQue, HotpotQA, 2Wiki) → label `'C'`
- Rationale: dataset construction **implies required reasoning depth**

Results and Discussion

Averaged over all datasets - GPT-3.5 (Turbo)

Type	Method	F1	Acc	Time
Simple	No Retrieval	48.56	44.27	0.71
	Single-Step Approach	46.99	45.27	1.00
Adaptive	Adaptive Retrieval	48.20	45.30	0.86
	Self-RAG*	22.98	34.13	1.50
	Adaptive-RAG (Ours)	**50.91**	**48.97**	**1.46**
Complex	Multi-Step Approach	50.87	49.70	3.33
Oracle	Adaptive-RAG w/ Oracle	62.80	58.57	1.03

*Self-RAG uses a different base LLM (LLaMA2-13B)

Asai et al., ICLR 2024

Adapting to Query Complexity

Simple vs. Complex Datasets

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Simple Dataset → Natural Questions

Method	F1	Time
No Retrieval	19.00	0.13×
Single-step	47.30	1.00×
Adaptive Retrieval	36.00	0.56×
Adaptive-RAG (Ours)	**47.30**	1.00×
Multi-step	47.80	10.18×

✓ *Matches peak accuracy at minimal cost*

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Complex Dataset → HotpotQA

Method	F1	Time
No Retrieval	22.71	0.11×
Single-step	46.15	1.00×
Adaptive Retrieval	32.22	0.55×
Adaptive-RAG (Ours)	**53.82**	5.99×
Multi-step	56.54	9.38×

✓ *Detects complexity, boosts F1 by +21.6 pts*

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"When the question is easy, we act fast. ***When the question is hard, we reason deep."***

Classifier Performance

Confusion Matrix

Confusion Matrix for Adaptive-RAG Classifier

Actual ↓ / Predicted →	No	One	Multi
No Retrieval	**31%**	47%	22%
Single-step	10%	**66%**	23%
Multi-step	3%	31%	**65%**

Diagonal (highlighted) = correct predictions · off-diagonal = misclassifications

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- Classifier performs well on **One** (66%) and **Multi** (65%)
- Critical risk: **Multi** misclassified as **One** (31%), leads to wrong answers

Classifier Size Analysis

Performance is robust across all model sizes

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T5-small (60M params)	T5 Base - 223M params	T5 Large - 770M params
QA Performance	QA Performance	QA Performance
F1: 45.83	**F1: 45.97**	**F1: 46.94**
Steps / Classifier Acc	Steps / Classifier Acc	Steps / Classifier Acc
964 steps	**983 steps**	**1084 steps**
53.48% acc	**53.41% acc**	**54.52% acc**

*F1 scores remain stable across all sizes (**45.83** → **46.94**)* · Smaller classifiers are viable for **resource-constrained** deployments

Case Study 1

Single-hop (Natural Questions)

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Question: Which famous logo changed to a flat sans-serif font since 1999?

Adaptive Retrieval (Single-step)

Unnecessarily fetches documents → misleading context

Answer: **Microsoft** *(wrong)*

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Adaptive-RAG, Type A (No Retrieval)

LLM's own knowledge is sufficient

Answer: **Google** *(correct)*

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LLM's own knowledge is sufficient

Answer: **Google** *(correct)*

- Simple query, no retrieval needed
- **Adaptive-RAG** avoids unnecessary fetch and answers correctly

Case Study 2

Multi-hop (MuSiQue)

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Question: Who is the child of the Italian navigator who explored the coast where César Gaytan was born?

Adaptive Retrieval (No Retrieval)

Skips retrieval for a complex query

Answer: **Giovanni Caboto** *(wrong)*

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Detects complexity, retrieves iteratively

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Adaptive-RAG, Type C (Multi-step)

Detects complexity, retrieves iteratively

Answer: **Sebastian Cabot** *(correct)*

- Complex query, multi-step reasoning needed
- **Adaptive-RAG** detects complexity and retrieves iteratively to find the correct answer

Summary & Conclusion

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- **Accuracy preserved at lower cost:** matches multi-step F1 using fewer steps
- **No human labelling required:** classifier trained fully automatically
- **Modular and plug-and-play:** works with any LLM and retriever out of the box
- **Tiny classifier is sufficient:** 60M params performs on par with 770M T5-Large

Limitations & Research Gaps

- **Only 3 complexity tiers:** finer-grained complexity levels unexplored
- **Oracle classifier gap:** classifier F1 50.91 vs oracle 62.80 significant room to improve
- **Auto-labelling noise:** incorrect labels may hurt classifier generalization

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Thank you for your attention!



Paper: Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity

Jeong et al., 2024 arXiv:2403.14403

Back-up Slides, for Q&A

Ablation: Which Labelling Strategy Works Best?

Strategy	F1	Steps	Classifier Acc
Full (Silver + Bias)	**46.94**	**1,084**	**54.52%**
Silver only (no bias)	48.79	**1,464** ↑	40.00%
Bias only (no silver)	43.43	640	60.30%

Key Takeaway

- Silver-only gives higher F1 but far more steps (1,464 vs 1,084)
- Bias-only alone loses F1 score
- **Both strategies together achieve the best efficiency–accuracy trade-off**

Classifier Performance & Time Savings

Query Distribution and Time per Type

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Type A - No Retrieval	Type B - Single-step	Type C - Multi-step
% of Queries	% of Queries	% of Queries
8.6%	**53.33%**	**38.07%**
Time per Query	Time per Query	Time per Query
0.35s	**3.08s**	**27.18s**

*Most queries (53%) need only a **single step**, only 38% require expensive **multi-step** reasoning*

The Retriever: BM25

The Search Engine for Facts

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What is BM25?

- **Keyword Searcher**: It acts like a mini Google for the system.
- **Fact Finder**: It scans millions of Wikipedia pages to find sentences that match the user's question.
- **Input Provider**: It gives the LLM the open-book text it needs to find the correct answer.

Robertson et al., TREC-3 1994

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Why is it used in this study?

- **Fair Comparison**: Every model (Simple or Complex) uses the *exact same* BM25 tool so the test is fair.
- **On-Demand**: It only runs when the **Classifier** decides the LLM isn't smart enough to answer alone (Labels B or C).

Robertson et al., TREC-3 1994